

## Evaluation Metrics for Classification

We have the following commonly used evaluation metrics that helps us assess the performance of our model.

- 1. Confusion Matrix:** It's a table used to evaluate the performance of a classification model by comparing actual labels with predicted labels.

|        |          | Predicted |          |
|--------|----------|-----------|----------|
|        |          | Positive  | Negative |
| Actual | Positive | TP        | FN       |
|        | Negative | FP        | TN       |

In Confusion Matrix (CM):

- **True Positive (TP):** Model correctly predicts the positive class
- **True Negative (TN):** Model correctly predicts the negative class
- **False Positive (FP):** Model predicts positive but actual is negative (Type I error)
- **False Negative (FN):** Model predicts negative but actual is positive (Type II error)

Let's take an **example** of a model predicting if a person is sick or not, the CM looks something like this:

|                  |          | ACTUAL VALUES |             |
|------------------|----------|---------------|-------------|
|                  |          | Positive      | Negative    |
| PREDICTED VALUES | Positive | TP<br>(30)    | FP<br>(30)  |
|                  | Negative | FN<br>(10)    | TN<br>(930) |

Sick People correctly predicted as sick by the model (TP)  
 Healthy people incorrectly predicted as sick by the model (FP)  
 Sick people incorrectly predicted as not sick by the model (FN)  
 Healthy people correctly predicted as not sick by the model (TN)

## Why does CM matter?

Measuring accuracy alone for a model can be misleading. So, CM shows what types of errors the model is making.

- **Type I Error** Is more important when false alarms are costly or dangerous, so we try to avoid FP.
  - *Examples* - email spam detection, loan approval, judicial system etc.
  - **Precision** controls Type I Errors
- **Type II Error** Is more important when missing a positive case is costly or dangerous, so we try to avoid FN.
  - *Examples* - medical diagnosis, fraud detection, fire alarms etc.
  - **Recall** controls Type II Errors

1. **Accuracy: Accuracy** provides the proportion of correctly classified instances.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total}}$$

2. **Precision: Precision** focuses on the accuracy of positive predictions.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

3. **Recall (Sensitivity or True Positive Rate): Recall** measures the proportion of correctly predicted positive instances among all actual positive instances.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

4. **F1 Score: F1 score** is the harmonic mean of precision and recall.

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

| *Keep Learning & Keep Exploring!*

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